

Mg TRINITY DN1546 c0 g1 serine rm 1	TPGAKTDSGDGVD.....SIVLPSTPKGSGPGF.	28
Mg TRINITY DN1546 c0 g1 serine rm 2	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPGF.	76
Mg TRINITY DN1546 c0 g1 serine rm 3	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPAF.	76
Mg TRINITY DN1546 c0 g1 serine rm 4	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPGF.	76
Mlaby 7048 AgSp1 serine rm1a 1	PGTTPGAETGPDGNVD.....KIILPTTPVRPEPQQ	32
Mlaby 7048 AgSp1 serine rm1a 3	PGTTPGVVTGPDGQP.V.QFIVPQGAFTTPGSIPGPNKPVH.VRPAGPGTTPGAKTDSGSIQ.....SIVLPSTPKPSGPGF.	76
Mlaby 7048 AgSp1 serine rm1a 12	FIVPQGAFTTPGSIPGPKGKPVH.VRPAGPGTTPGAKTDSGNID.....SIVLPSTPKPSRPGK.	59
Mlaby 7048 AgSp1 serine rm1a 14	FIVPQGAFTTPGSIPGPDGKQVH.VRPAGPGTTPGAK.....	36
Mlaby 7048 AgSp1 serine rm1a 25		0
Mlaby 7048 AgSp1 serine rm1a 26	PGTTPGVVTGPGGKPL.R.....	17
Mlaby 7048 AgSp1 serine rm1a 27		0
Mlaby 7048 AgSp1 serine rm1a 28	PGTTPGTVTGPDGNP.VKFIVPQGAFTTPGSISGPNKPIH.VRPALPGTTPGAK.....	53
Mlaby 7048 AgSp1 serine rm1a 37		0
Mlaby 7048 AgSp1 serine rm1a 38	PGTTPGVVTGPGGQPL.RFIVPQGAFTTPGSIPGPNKQPIH.VRPAGPGTTPGAKTDSGNID.....SIVLPSTPKGSGPVP.	76
Msag 19329 AgSp1.1 serine rm 1	AFTTPGSIAGPHGKRVH.VKPAGPELHQE.LKPSDGDVS.....SIVLPSTPKGSG.PG.	51
Msag 19329 AgSp1.1 serine rm 2	PGTTPGVVTGPDGQP.VKYVVPQ.EPYDSWIYCRTSRKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPKGSG.PG.	74
Msag 19329 AgSp1.1 serine rm 3	PGTTPGVVTGPDGQP.VKHVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPKASG.PG.	75
Msag 19329 AgSp1.1 serine rm 4	PGTTPGVVTGPDGQP.VKYVVPQVAFSTTPGSIAGPHGKRVHAREPAGPGTTPGAKTDSGDVS.....SIVLPSTPKGSG.PG.	76
Msag 19329 AgSp1.1 serine rm 5	PGTTPGVVTGPDGQSQFKYVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDPDGDVS.....SIVLPSTPKGSG.PG.	76
Msag 19329 AgSp1.1 serine rm 6	PGTTPGVVTGPDGQP.VKYVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPK.....	70
Ncruc 58532and13532 AgSp1 serine rm 87	SPETSNG	7
Ncruc 58532and13532 AgSp1 serine rm 88	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VKPAGPGTTPGAKTDSGSDID.....SIVLPSTPK.....	70
Ncruc 58532and13532 AgSp1 serine rm 89	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VKPAGPGTTPGAKTNSDGSID.....SIVLPSTPK.....	70
Ncruc 58532and13532 AgSp1 serine rm 90	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VRPAGPGTTPGAKTDSGSDID.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 91	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VGPAGPGTTPGAKTDSGSDID.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 92	PGTTPGVVTGPDGQP.VKFIVPQGAFTSTPGTIPGPHGKPIH.IGPAGPGTTPGAKXDSGSDID.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 93	PGTTPGVVTGPDGQP.VKFFVPNGAFSTPGSIPGPHGKPIH.IGPAGPGTTPGAKLDSGDIDIS.....EILLPSTP.SP.....	71
Lcorn 58650 AgSp1.2 serine rm1 3		0
Lcorn 58650 AgSp1.2 serine rm1 4	PGTTPGVVTGPDGQP.VKFIVPQGAFTTPGSIPGPNKPIH.VKPAGPGTTPGAKTNSDGSIDSIV.....	64
Lcorn 58650 AgSp1.2 serine rm1 5	PGTTPGVVTGADGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 6	FIVPQGAFTTSGSIPGPNKPIH.VKPAGPGTTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 7		0
Lcorn 58650 AgSp1.2 serine rm1 8	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 10		0
Lcorn 58650 AgSp1.2 serine rm1 11	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 12	FIVPQGAFTTPGSILGPNKPIH.VKPAGPGTTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 17	FIVPQGAFTTPGSIPGPNKPIR.....	23
Lcorn 58650 AgSp1.2 serine rm1 21		0
Lcorn 58650 AgSp1.2 serine rm1 22	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 45	TNSDGNID.....SIVLPSTPK.....	17
Lcorn 58650 AgSp1.2 serine rm1 56		6
Lcorn 58650 AgSp1.2 serine rm1 57	PGTSPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 69		0
Lcorn 58650 AgSp1.2 serine rm1 70	PGTSPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 75		0
Lcorn 58650 AgSp1.2 serine rm1 76		0
Lcorn 58650 AgSp1.2 serine rm1 77	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 79	IDSNGSID.....SIVLPSTPKGSGP...	21
Lcorn 58650 AgSp1.2 serine rm1 80	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 82		0
Lcorn 58650 AgSp1.2 serine rm1 83	PGTTPGVVTGPDGQPVK.FIVPNGAFSTPGSIPGPNKPIH.VGPAG.....	45
Aarg AgSp1 FC511 serine rm1 14	PGTTPGTMTGSDGKPK.NKFVVPKGAFTTPGSFPGPDGKPIH.VEPAGPGSTPGTGTGPDGSIK.....QVFPVPTPKSSESSGG	77

Mg TRINITY DN1546 c0 g1 serine rm 1	QS.....	PTTPSHVTKPDGQPIQIVP	49
Mg TRINITY DN1546 c0 g1 serine rm 2	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mg TRINITY DN1546 c0 g1 serine rm 3	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mg TRINITY DN1546 c0 g1 serine rm 4	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mlaby 7048 AgSp1 serine rm1a 1	N.....	TTTKVKGPQGKPLVVP	50
Mlaby 7048 AgSp1 serine rm1a 3	N.....	FQTPSPVTKPDGQPIQIVP	96
Mlaby 7048 AgSp1 serine rm1a 12	N.....	QFKNPLPVTQSDGQPIQIVP	79
Mlaby 7048 AgSp1 serine rm1a 14	N.....	NPLPVTQSDGQPIQIVP	36
Mlaby 7048 AgSp1 serine rm1a 25	N.....	NPLPVTQSDGQPIQIVP	17
Mlaby 7048 AgSp1 serine rm1a 26	N.....	NPLPVTQSDGQPIQIVP	17
Mlaby 7048 AgSp1 serine rm1a 27	GPGGQPRPQSPG.....	QQTTPTEPTPGPMCKPLQIVP	33
Mlaby 7048 AgSp1 serine rm1a 28	N.....	NPLPVTQPDGQPIQIVP	53
Mlaby 7048 AgSp1 serine rm1a 37	ITT.....	ITTTLPITPGPGPEPLQIVP	17
Mlaby 7048 AgSp1 serine rm1a 38	ITT.....	ITTTLPITPGPGPEPLQIVP	94
Msag 19329 AgSp1.1 serine rm 1	F.....	QSPTPSQVTKPDGQPIQIVP	72
Msag 19329 AgSp1.1 serine rm 2	F.....	HTPTPSQVTKPDGQPIQIVP	95
Msag 19329 AgSp1.1 serine rm 3	F.....	QSPTPSQVTKPDGQPIQIVP	96
Msag 19329 AgSp1.1 serine rm 4	F.....	RTPTLSQVTKPDGQPIQIVP	97
Msag 19329 AgSp1.1 serine rm 5	F.....	QSPTPSQVTKPDGQPIQIVP	97
Msag 19329 AgSp1.1 serine rm 6	F.....	QSPTPSQVTKPDGQPIQIVP	70
Ncruc 58532and13532 AgSp1 serine rm 87	K.....KGTGENPQNANNSLGPLQPDSTTTPGSQPGNPMGPGSGPGGQTTPGSQGPVDFD...	HTPSPITKPDGKPIQIVP	79
Ncruc 58532and13532 AgSp1 serine rm 88	GSGPGFQF...	QTPQPVTKPDGQPIQIVP	96
Ncruc 58532and13532 AgSp1 serine rm 89	GSGPGFQF...	QTPQPVTKPDGQPIQIVP	96
Ncruc 58532and13532 AgSp1 serine rm 90	SGPGFPPTPKGSGPGFQF...	QTPQPVTKPDGQPIQIVP	107
Ncruc 58532and13532 AgSp1 serine rm 91	SG.....PGFNF...	ETPRPAPQSGGPIQIVP	95
Ncruc 58532and13532 AgSp1 serine rm 92	SG.....PGYNF...	ETPRPAPQSGGPIQIVP	95
Ncruc 58532and13532 AgSp1 serine rm 93	PAP.....QPA.....	NPTTPTDVRGPQCSPIPIVP	97
Lcorn 58650 AgSp1.2 serine rm1 3	GSGPG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 4	LPSTPKGSRPGFQF...	QTPQPIITKSGGQPIQIVP	96
Lcorn 58650 AgSp1.2 serine rm1 5	LPSTPKGSRPGFQF...	QTPQPIITKSGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 6	LPSTPKGSRPGFQF...	QTPQPIITKSGGQPIQIVP	36
Lcorn 58650 AgSp1.2 serine rm1 7	LPSTPKGSRPGFQF...	QTPQPIITKSGGQPIQIVP	10
Lcorn 58650 AgSp1.2 serine rm1 8	LPSTPKGSRPGFQF...	QTPQPIITKSGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 10	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 11	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 12	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	36
Lcorn 58650 AgSp1.2 serine rm1 17	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	23
Lcorn 58650 AgSp1.2 serine rm1 21	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	10
Lcorn 58650 AgSp1.2 serine rm1 22	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 45	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 56	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 57	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 69	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 70	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 75	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	11
Lcorn 58650 AgSp1.2 serine rm1 76	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 77	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 79	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	43
Lcorn 58650 AgSp1.2 serine rm1 80	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 82	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 83	GSGTG.....	FQFQTPQPIITKPGGQPIQIVP	45
Aarg AgSp1 FC511 serine rm1 14	KKGTGENPANSNNAIGPLQT.....	TSTTTPTSISGPDGQPMQVVP	118

Mg TRINITY DN1546 c0 g1 serine rm 1	AG	51
Mg TRINITY DN1546 c0 g1 serine rm 2	AG	99
Mg TRINITY DN1546 c0 g1 serine rm 3	AG	99
Mg TRINITY DN1546 c0 g1 serine rm 4	AG	99
Mlaby 7048 AgSp1 serine rm1a 1	AG	52
Mlaby 7048 AgSp1 serine rm1a 3	AG	98
Mlaby 7048 AgSp1 serine rm1a 12	AG	81
Mlaby 7048 AgSp1 serine rm1a 14	..	36
Mlaby 7048 AgSp1 serine rm1a 25	AG	19
Mlaby 7048 AgSp1 serine rm1a 26	..	17
Mlaby 7048 AgSp1 serine rm1a 27	AG	35
Mlaby 7048 AgSp1 serine rm1a 28	..	53
Mlaby 7048 AgSp1 serine rm1a 37	AG	19
Mlaby 7048 AgSp1 serine rm1a 38	AG	96
Msag 19329 AgSp1.1 serine rm 1	AG	74
Msag 19329 AgSp1.1 serine rm 2	AG	97
Msag 19329 AgSp1.1 serine rm 3	AG	98
Msag 19329 AgSp1.1 serine rm 4	AG	99
Msag 19329 AgSp1.1 serine rm 5	AG	99
Msag 19329 AgSp1.1 serine rm 6	..	70
Ncruc 58532and13532 AgSp1 serine rm 87	AG	81
Ncruc 58532and13532 AgSp1 serine rm 88	AG	98
Ncruc 58532and13532 AgSp1 serine rm 89	AG	98
Ncruc 58532and13532 AgSp1 serine rm 90	AG	109
Ncruc 58532and13532 AgSp1 serine rm 91	AG	97
Ncruc 58532and13532 AgSp1 serine rm 92	AG	97
Ncruc 58532and13532 AgSp1 serine rm 93	AG	99
Lcorn 58650 AgSp1.2 serine rm1 3	AG	28
Lcorn 58650 AgSp1.2 serine rm1 4	AG	98
Lcorn 58650 AgSp1.2 serine rm1 5	..	17
Lcorn 58650 AgSp1.2 serine rm1 6	..	36
Lcorn 58650 AgSp1.2 serine rm1 7	AG	12
Lcorn 58650 AgSp1.2 serine rm1 8	..	17
Lcorn 58650 AgSp1.2 serine rm1 10	AG	28
Lcorn 58650 AgSp1.2 serine rm1 11	..	17
Lcorn 58650 AgSp1.2 serine rm1 12	..	36
Lcorn 58650 AgSp1.2 serine rm1 17	..	23
Lcorn 58650 AgSp1.2 serine rm1 21	AG	12
Lcorn 58650 AgSp1.2 serine rm1 22	..	17
Lcorn 58650 AgSp1.2 serine rm1 45	..	17
Lcorn 58650 AgSp1.2 serine rm1 56	AG	28
Lcorn 58650 AgSp1.2 serine rm1 57	..	17
Lcorn 58650 AgSp1.2 serine rm1 69	AG	28
Lcorn 58650 AgSp1.2 serine rm1 70	..	17
Lcorn 58650 AgSp1.2 serine rm1 75	..	11
Lcorn 58650 AgSp1.2 serine rm1 76	AG	28
Lcorn 58650 AgSp1.2 serine rm1 77	..	17
Lcorn 58650 AgSp1.2 serine rm1 79	AG	45
Lcorn 58650 AgSp1.2 serine rm1 80	..	17
Lcorn 58650 AgSp1.2 serine rm1 82	AG	28
Lcorn 58650 AgSp1.2 serine rm1 83	..	45
Aarg AgSp1 FC511 serine rm1 14	AG	120

 non-conserved
 similar
 $\geq 50\%$ conserved